

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C03: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

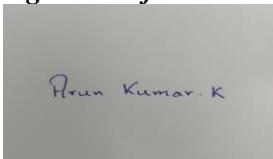
Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I Kurangi Arun Kumar/192372287 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Answer:

Got it! I will avoid using standard LaTeX syntax ($\$or\$$) for formatting labels, text headers, or plain math so no dollar signs appear in your view.

Here is Case Study 1 rendered cleanly in standard text:

Case Study 1 — Smart Medical Diagnosis System

1. Propositional Logic Representation

Symbols

- **F**: Patient A has Fever
- **C**: Patient A has Cough
- **R**: Patient A has Rash
- **B**: Patient A has Breathlessness
- **Flu**: Patient A has Possible Flu
- **M**: Patient A has Possible Measles
- **P**: Patient A has Possible Pneumonia

Knowledge Base (KB)

- **Rules:**
 - R1: $(F \wedge C) \rightarrow \text{Flu}$
 - R2: $(F \wedge R) \rightarrow M$
 - R3: $(C \wedge B) \rightarrow P$
- **Facts:**

- F1: F = True
- F2: C = True
- F3: B = True

2. Forward Chaining

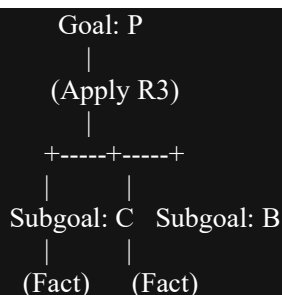
Forward chaining begins with known facts and applies Modus Ponens to derive new facts iteratively.

- **Initial Known Facts:** {F, C, B}
- **Step 1:**
 - Check R1: Antecedent ($F \wedge C$) is satisfied because F and C are in the known facts set.
 - **Rule Fired:** R1
 - **Fact Derived:** Flu
 - **Updated Known Facts:** {F, C, B, Flu}
- **Step 2:**
 - Check R2: Antecedent ($F \wedge R$) is not satisfied (R is missing).
 - Check R3: Antecedent ($C \wedge B$) is satisfied because C and B are in the known facts set.
 - **Rule Fired:** R3
 - **Fact Derived:** P
 - **Updated Known Facts:** {F, C, B, Flu, P}

Derived Diagnoses for Patient A: Possible Flu (Flu) and Possible Pneumonia (P).

3. Backward Chaining & Goal Reduction Tree

- **Goal:** P (Patient A has Pneumonia)



1. **Goal:** Prove P.
2. Match P with the consequent of R3: $(C \wedge B) \rightarrow P$.
3. Set subgoals: Prove C and prove B.
 - a. Subgoal C matches Fact F2 ($C = \text{True}$).
 - b. Subgoal B matches Fact F3 ($B = \text{True}$).
4. Both subgoals succeed, verifying the goal **P = True**.

Question 2 – Student Assignment Submission Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Answer :

1. Propositional Logic Representation

Symbols

- **S**: A student submits the assignment
- **M**: A student receives internal marks

Knowledge Base (KB)

- **Premise 1 (Rule):** $S \rightarrow M$ (If a student submits the assignment, then the student receives internal marks)
- **Premise 2 (Fact):** S (Rahul submitted the assignment)

Goal

- Prove **M** (Rahul receives internal marks)

2. CNF Conversion & Goal Negation

Premise Transformations to CNF

- **For Premise 1 ($S \rightarrow M$):**
 - Eliminate implication ($A \rightarrow B \equiv \neg A \vee B$):

Clause 1 (C1): $\neg S \vee M$

- **For Premise 2 (S):**

- Already an atomic literal:

Clause 2 (C2): S

Negation of Goal

- **Goal:** M
- **Negated Goal:** $\neg M$

Clause 3 (C3): $\neg M$

3. Step-by-Step Resolution Algorithm

We combine all clauses $\{C1, C2, C3\}$ into a single set and resolve to derive a contradiction.

1. Initial Set of Clauses:

- a. $C1: \neg S \vee M$
- b. $C2: S$
- c. $C3: \neg M$

2. Step 1: Resolve $C1 (\neg S \vee M)$ and $C2 (S)$ on literal **S**:

- a. The complementary literals S and $\neg S$ cancel out.
- b. **Derived Clause (C4):** M

3. Step 2: Resolve $C4 (M)$ and $C3 (\neg M)$ on literal **M**:

- a. The complementary literals M and $\neg M$ cancel out.
- b. **Derived Clause (C5):** $\square$ (Empty Clause)

4. Resolution Proof Tree

```
(C1)  $\neg S \vee M$       (C2)  $S$ 
-----
(C4)  $M$               (C3)  $\neg M$ 
-----
(C5)  $\square$ 
```

5. Conclusion

Since the **Empty Clause** ($\square$) is successfully derived, a logical contradiction has been established with the negated goal. Therefore, the original goal is proved: **Rahul receives internal marks ($M = \text{True}$)**.

Question 3 – Library Membership

Knowledge Base

- 1. If a person is a library member, then the person can borrow books.
- 2. Priya is a library member.

Goal

- 1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

- 1. Represent the knowledge base using propositional logic.
- 2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Answer :

1. Propositional Logic Representation

Symbols

- **L:** A person is a library member (Priya is a library member)
- **B:** A person can borrow books (Priya can borrow books)

Knowledge Base (KB)

- **Premise 1 (Rule):** $L \rightarrow B$ (If a person is a library member, then the person can borrow books)
- **Premise 2 (Fact):** L (Priya is a library member)

Goal

- Prove **B** (Priya can borrow books)

2. CNF Conversion & Goal Negation

Premise Transformations to CNF

- **For Premise 1 ($L \rightarrow B$):**
 - Eliminate implication ($A \rightarrow B \equiv \neg A \vee B$):

Clause 1 (C1): $\neg L \vee B$

- **For Premise 2 (L):**
 - Already an atomic literal:

Clause 2 (C2): L

Negation of Goal

- **Goal:** B
- **Negated Goal:** $\neg B$

Clause 3 (C3): $\neg B$

3. Step-by-Step Resolution Algorithm

We combine all clauses $\{C1, C2, C3\}$ into a single set and apply resolution to derive a contradiction.

1. Initial Set of Clauses:

- a. $C1: \neg L \vee B$
- b. $C2: L$
- c. $C3: \neg B$

2. Step 1: Resolve C1 ($\neg L \vee B$) and C2 (L) on literal **L**:

- a. The complementary literals L and $\neg L$ cancel out.
- b. **Derived Clause (C4):** B

3. **Step 2:** Resolve C4 (B) and C3 ($\neg B$) on literal **B**:
 - a. The complementary literals B and $\neg B$ cancel out.
 - b. **Derived Clause (C5):** $\square$ (Empty Clause)

4. Resolution Proof Tree

(C1) $\neg L \vee B$	(C2) L
(C4) B	(C3) $\neg B$
(C5) $\square$	

5. Final Conclusion

Since the **Empty Clause** ($\square$) is successfully derived, a logical contradiction has been reached with the negated goal. Therefore, the goal is proved: **Priya can borrow books (B = True)**.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Answer :

1. Propositional Logic Representation

Symbols

- **A:** A student clears the aptitude test (Arun cleared the aptitude test)
- **E:** A student is eligible for placement (Arun is eligible for placement)

Knowledge Base (KB)

- **Premise 1 (Rule):** $A \rightarrow E$ (If a student clears the aptitude test, then the student is eligible for placement)
- **Premise 2 (Fact):** A (Arun cleared the aptitude test)

Goal

- Prove E (Arun is eligible for placement)

2. CNF Conversion & Goal Negation

Premise Transformations to CNF

- **For Premise 1 ($A \rightarrow E$):**

- Eliminate implication ($X \rightarrow Y \equiv \neg X \vee Y$):

Clause 1 (C1): $\neg A \vee E$

- **For Premise 2 (A):**

- Already an atomic literal:

Clause 2 (C2): A

Negation of Goal

- **Goal:** E
- **Negated Goal:** $\neg E$

Clause 3 (C3): $\neg E$

3. Step-by-Step Resolution Algorithm

We combine all clauses $\{C1, C2, C3\}$ into a single set and apply resolution to derive a contradiction.

1. Initial Set of Clauses:

- a. C1: $\neg A \vee E$
- b. C2: A
- c. C3: $\neg E$

2. Step 1: Resolve C1 ($\neg A \vee E$) and C2 (A) on literal A:

- a. The complementary literals A and $\neg A$ cancel out.
- b. **Derived Clause (C4):** E

3. Step 2: Resolve C4 (E) and C3 ($\neg E$) on literal E:

- a. The complementary literals E and $\neg E$ cancel out.
- b. **Derived Clause (C5):** $\square$ (Empty Clause)

4. Resolution Proof Tree

```
(C1)  $\neg A \vee E$       (C2) A
-----
(C4) E                (C3)  $\neg E$ 
-----
(C5)  $\square$ 
```

5. Conclusion

Since the **Empty Clause** ($\square$) is successfully derived, a logical contradiction has been established with the negated goal. Therefore, the goal is proved: **Arun is eligible for placement (E = True)**.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

1. Propositional Logic Representation

Symbols

- **P:** The user enters the correct password
- **A:** The user is authenticated
- **G:** The user is granted access

Knowledge Base (KB)

- **Premise 1 (Rule 1):** $P \rightarrow A$ (If a user enters the correct password, then the user is authenticated)
- **Premise 2 (Rule 2):** $A \rightarrow G$ (If a user is authenticated, then the user is granted access)
- **Premise 3 (Fact):** P (The user entered the correct password)

Goal

- Prove **G** (The user is granted access)

2. CNF Conversion & Goal Negation

Premise Transformations to CNF

- **For Premise 1 ($P \rightarrow A$):**
 - Eliminate implication ($X \rightarrow Y \equiv \neg X \vee Y$):

Clause 1 (C1): $\neg P \vee A$

- **For Premise 2 ($A \rightarrow G$):**
 - Eliminate implication ($X \rightarrow Y \equiv \neg X \vee Y$):

Clause 2 (C2): $\neg A \vee G$

- **For Premise 3 (P):**
 - Already an atomic literal:

Clause 3 (C3): P

Negation of Goal

- **Goal:** G
- **Negated Goal:** $\neg G$

Clause 4 (C4): $\neg G$

3. Step-by-Step Resolution Algorithm

We combine all clauses $\{C1, C2, C3, C4\}$ into a single set and resolve to derive the empty clause.

1. Initial Set of Clauses:

- a. $C1: \neg P \vee A$
- b. $C2: \neg A \vee G$
- c. $C3: P$
- d. $C4: \neg G$

2. **Step 1:** Resolve C1 ($\neg P \vee A$) and C3 (P) on literal **P**:

a. The complementary literals P and $\neg P$ cancel out.

b. **Derived Clause (C5):** A

3. **Step 2:** Resolve C2 ($\neg A \vee G$) and C5 (A) on literal **A**:

a. The complementary literals A and $\neg A$ cancel out.

b. **Derived Clause (C6):** G

4. **Step 3:** Resolve C6 (G) and C4 ($\neg G$) on literal **G**:

a. The complementary literals G and $\neg G$ cancel out.

b. **Derived Clause (C7):** $\square$ (Empty Clause)

4. Resolution Proof Tree

(C1) $\neg P \vee A$	(C3) P

(C5) A	(C2) $\neg A \vee G$

(C6) G	(C4) $\neg G$

(C7) $\square$	

5. Final Conclusion

Since the **Empty Clause** ($\square$) is successfully derived, a logical contradiction has been established with the negated goal. Therefore, the original goal is proved: **The user is granted access (G = True)**.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation